Ex.No. : 05

Capturing and Analyzing the Packet Traces of UDP

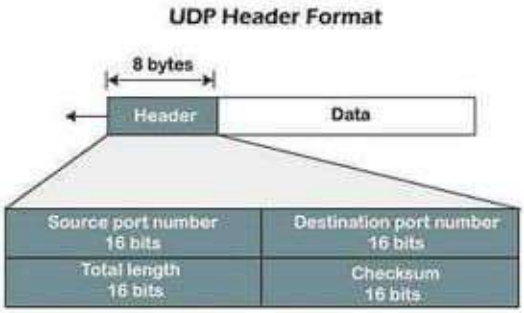
Aim

To capture and analyze the packet traces of UDP protocol using Wireshark

UDP

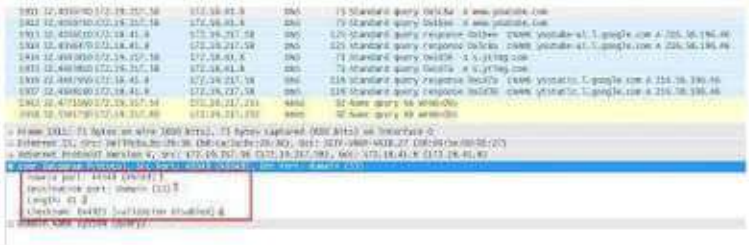
- The User Datagram Protocol (UDP) is a lightweight data transport protocol that works on top of IP.
- UDP provides a mechanism to detect corrupt data in packets, but it does not attempt to solve other problems that arise with packets, such as lost or out of order packets. That's why UDP is sometimes known as the Unreliable Data Protocol.
- UDP is simple but fast, at least in comparison to other protocols that work over IP. It's often used for time-sensitive applications (such as real-time video streaming) where speed is more important than accuracy.

UDP Packet structure



1. Select one UDP packet from your trace. From this packet, determine how many fields there are in the UDP header.
   Answer: There are four fields in the header: source port, destination port, Length, and checksum.

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2. The value in the Length field is the length of what? Verify your claim with your captured UDP packet.
   Answer: The UDP length field is the length of the header and data fields of the UDP segment, measured in bytes)." The displayed packet has a length field of 41 bytes. We know there are 8 byres of header. If we look at the packet content field, we also find 33 bytes of hexadecimal or ASCII-encoded data which corresponds to the payload of this UDP segment.
3. What is the maximum number of bytes that can be included in a UDP payload?
   Answer: Since there are only 16 bits, the maximum length of a UDP segment (including header) is 2^16 -1 or 65535 bytes.
4. What is the largest possible source port number?
   Answer: Since there are only 16 bits, the maximum source port number is 2^16 -1 or 65535 bytes
5. What is the protocol number for UDP? Give your answer in both hexadecimal and decimal notation.
   Answer : By looking the Protocol field of the IP datagram containing this UDP segment , UDP has a protocol number of 17 (this number is displayed in Wireshark as the value of the "protocol:" field in the IPV4 datagram.)
6. Examine a pair of UDP packets in which the first packet is sent by your host and the second packet is a reply to the first packet. Describe the relationship between the port numbers in the two packets.
   Answer: Let's look at packets query and response of NDS in the trace. These packets carry DNS application-level messages encapsulated inside of UDP. The IP address of the sender of query packet is the IP address of the destination of response packet , and the IP address of the destination of query packet is the IP address of the sender of response packet 2. The value of the source port in query packet is the value of the destination port of response packet; the value of the destination port of query packet is the value of the source port of response packet.

Result:

Thus, capturing and analyzing UDP packets using Wireshark has been completed successfully.

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